



Eutrophication of marine ecosystems: explaining (non-)acceptance of governance approaches for reducing agricultural nutrient input

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Abstract

The Baltic Sea suffers from excessive input of nutrients leading to eutrophication as a key pressure for the marine ecosystem. Land-based pollution from agricultural activities constitutes a major source of nutrient input to marine environments. To reduce nutrient runoff from agriculture, European and national policymakers have introduced regulations, such as the EU-Nitrates Directive, the HELCOM-Baltic Sea Action Plan, and national fertilizer regulations. However, the effect of these regulations and measures has been limited so far. One reason for this is the limited acceptance of regulations by farmers. By combining theoretical-conceptual literature on individual acceptance with insights from empirical research on agri-environmental schemes (AES) and farmers' behavior, the paper develops a comprehensive analytical framework for systematically studying central determinants of individual acceptance of agri-environmental policy regulations. A local case study of a nearshore area at the Baltic coast in Northern Germany is conducted, where intensive agricultural production takes place. The paper uses qualitative content analysis of twenty semi-structured qualitative interviews with local farmers as well as political and administrative authorities in the region. Findings indicate that, in particular, economic concerns, policy design factors, and different dimensions of political mistrust – interacting in a wicked-problem complex – cause substantial acceptance deficits. Apart from providing new conceptual insights for multi-disciplinary research on individual acceptance of (marine) environmental protection, the paper also offers insights about how to further develop governance approaches for reducing agricultural nutrient input in marine water, as well as for other land-based sources of marine pollution.

Keywords Marine pollution · Political regulation · Agricultural sector · Acceptance · Land-sea interaction · Baltic sea

Introduction

The Baltic Sea is heavily affected by excessive input of nitrogen. While the situation has improved in recent years, at least 93.8% of the marine area still is below good environmental status for eutrophication according to the latest HELCOM assessment report (HELCOM, 2023). Eutrophication is mainly driven by wastewater and agricultural inputs as well as nutrient leakages from the sediments (HELCOM, 2023). Agricultural activities constitute the most significant source of nitrate input to marine areas, posing a threat to the entire marine ecosystem by causing excessive blooms

of algae, increased water turbidity, and eventually oxygen depletion with fatal implications for marine benthic communities and – directly and indirectly – species at higher trophic levels such as fish, birds, and marine mammals (HELCOM, 2023). By degrading the water quality, eutrophication not only affects the marine ecosystem but may also significantly affect human health and economies. In response, we can find increasing governance measures in Europe to reduce nutrient input from the agricultural sector, such as the EU-Nitrates Directive, the HELCOM-Baltic Sea Action Plan, as well as national fertilizer ordinances. While nitrogen surplus trends have recently decreased in Germany, the country still has one of the highest nitrogen surpluses per hectare (HELCOM, 2023). One crucial explanation for the limited impact of the regulations and measures is the lack of acceptance of regulations on side of farmers (Michels et al. 2025).

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Existing empirical analyses on the determinants of farmers' decision-making exhibit several research gaps. In particular, determinants regarding the policy design of regulations have so far hardly been analyzed in detail. Moreover, there is a lack of comprehensive empirical insights on interacting mechanisms linking different explanatory factors of (non-) acceptance. To address these knowledge gaps, the present paper integrates existing insights from research in economics, sociology, psychology, and political science to develop and empirically apply a comprehensive analytical framework for systematically studying determinants of acceptance, and of acceptance deficits, of political regulations for reducing agricultural nutrient runoffs. The empirical case study focuses on the individual acceptance of the fertilizer ordinance in Schleswig–Holstein, a coastal federal state in Northern Germany, and is based on comprehensive semi-structured interviews with farmers. Insights of this paper are also of importance in the broader context of the current resistance of the agricultural sector to biodiversity protection and restoration in all of Europe (for example against the EU nature restoration law), which shows that (A) policy acceptance has to be taken into account when thinking about the effectivity of environmental and climate policy, and that (B) land-sea interaction is of crucial relevance in the sense that not only agricultural policies and activities influence the marine ecosystems, but also that actors from the agricultural sector have a crucial influence on marine policies.

Findings may finally offer useful policy-relevant insights about how to further develop binding and complementary non-binding policy regulations and enforcement mechanisms, for the case of agricultural nutrient runoff, as well as for other land-based sources of marine pollution such as pesticides or heavy metals, so as to overcome the challenges of political non-acceptance.

To demonstrate these contributions, the paper proceeds as follows: it first provides an overview of the state of the art of relevant theoretical-conceptual and empirical research in different disciplines, including persisting research gaps, followed by an introduction of the conceptual-analytical and methodological framework used for the empirical case study. The subsequent section presents the major results of the empirical analysis of determinants of (non-)acceptance of the fertilizer ordinance in Northern Germany. Results are discussed against the backdrop of previous research insights. The paper closes with some reflections on limitations of the methodological approach used in this paper and on avenues for future research.

Literature review and conceptual-theoretical framework

The concept of acceptance

Individual acceptance of regulations and norms is an object of research within various disciplines, such as sociology, economics, political science, or social psychology. The importance of individual acceptance of regulative frameworks has been widely recognized within research on agri-environmental policies (Schroeder et al. 2015; Massfeller et al. 2022) and beyond (Adell 2009; Huijts et al. 2012; Bug and Bukow 2016; Dermont et al. 2017). However, rather little work has been carried out regarding the conceptualization of an acceptance theory.

Still, there is some conceptual work, for example by Franken and Lenz (2004), Adell (2009), Huijts and Molin (2012), Dermont et al., (2017), and Massfeller et al., (2022). One of the most advanced conceptualizations of individual acceptance has been outlined by Franken and Lenz (2004) and was further elaborated by Huijts et al. (2012). The authors distinguish between attitudinal acceptance and behavioral acceptance. While attitudinal acceptance denotes that an existing regulative scheme or a system in general is supported by an individual in a cognitive way, behavioral acceptance is expressed by actual behavior, for example in concordance with a regulative scheme. Behavioral acceptance thus shows close similarity with the concept of compliance, which according to Young (1979) can be conceptualized as all behavior by subjects or actors that is conform to the requirements of behavioral prescriptions (Young 1979).

While attitudinal acceptance is often associated with behavioral acceptance or compliance (see, for example, Kraus 1995), the link between attitudes and behavior is not straightforward. For instance, research based on the theory of planned behavior has demonstrated that attitudes influence behavior rather indirectly rather than directly (Huijts and Molin, 2012; Ajzen 1991; see also Grelle and Hoffmann, 2024) and that there are various intervening variables, for example resource capacity.

Moreover, a high degree of behavioral acceptance might even be possible without attitudinal acceptance because of coercion that leads an actor to acting in accordance with behavioral prescriptions, even if he or she might not approve of it normatively (Franken and Lenz, 2004; Grelle and Hoffmann, 2024).

In this sense, the overall term of acceptance, as it is used in this paper, is broader than the concept of compliance. Consequently, existing research on determinants for explaining individual acceptance and non-acceptance, draws on compliance research literature but also considers

further explanatory determinants for explaining the attitudinal dimension of acceptance.

In this regard, in particular Huijts et al. (2012) have developed a comprehensive analytical framework for explaining technology acceptance, which can serve as a model for other policy areas. According to this framework, basic determinants of individual acceptance are, firstly, “gain motives”, assuming that peoples’ acceptance is dependent on perceived (economic) costs, risks, and benefits related to the adoption of technologies and behavior in concordance with regulatory schemes more in general. The authors hence draw on assumptions of neoclassical approaches of compliance research (Becker 1968; Herzfeld and Jongeneel 2012), arguing that compliance is more likely when the perceived costs of following the regulations are lower than the perceived costs of non-compliance. The latter depends on the perceived probability of being detected and the severity of any sanctions.

Secondly, based on assumptions of the normative activation theory (Schwartz and Howard 1981), “normative motives” are considered crucial for acceptance or non-acceptance in the framework of Huijts, Molin and Steg (Huijts et al. 2012; see also Schade and Schlag, 2003). Thirdly, affective responses such as positive or negative feelings towards the subject of the regulation are integrated into the framework. Furthermore, the authors underline that trust in actors that are responsible for the implementation of specific (regulative) actions increases individual acceptance of these actions. In addition, they consider fairness – including perceived fairness of the decision process of implementation as well as perceived fairness of the distribution of costs and benefits resulting from the implementation – as a crucial determinant of acceptance (referring to procedural and distributive fairness, see Huijts et al. 2012: 529). The authors, moreover, underline a close interdependency between trust and fairness perception- They find that a higher level of trust in responsible regulatory actors increases the probability of perceived greater fairness and that perceived greater fairness increases trust in responsible regulatory actors. Finally, the authors argue that knowledge, defined as thematic expertise, influences individual acceptance, as for example knowledge can influence people’s perception of related costs and benefits (see also Schade and Schlag, 2003). Moreover, it is assumed that little knowledge about details of a regulation rather fosters skepticism up to non-acceptance of this regulation.

While Adell’s (Adell 2009) more parsimonious analytical framework explaining technology acceptance also underlines the crucial relevance of the determinant of perceived individual gains in the sense of cost-benefits calculations, it moreover includes social influence as a further crucial explanatory for individual acceptance. It is argued

that opinions of important others (Adell 2009), such as family member and peers, contribute to explain the level of acceptance of systems and (political) decisions. The framework, moreover, also includes facilitating (organizational and technical) infrastructure as a third crucial determinant influencing individual acceptance, which may for example refer to specific policy design options of regulations.

Whereas the above-mentioned frameworks for explaining individual acceptance consider specific interaction between determinants of acceptance at least to some degree, Massfeller et al. (2022) compilation of determinants for individual acceptance of agri-environmental schemes (AES) by farmers refrains from doing so. Its strength and usefulness for the present paper, however, lies in the comprehensive consideration of insights of existing literature in the field of policy acceptance of AES.

According to Massfeller et al. (2022), a key determinant of acceptance is the perceived tradeoff between costs and benefits, which have also been stressed in the frameworks of Adell (2009) and Huijts et al. (2012), presented above. Furthermore, in line with the framework of Adell (2009), Massfeller et al. list social factors, such as other peoples’ (for example neighbors’) view of specific regulations and/or related farming practices, as influencing the individual acceptance of the regulation. Further crucial determinants according to the authors are moral or normative factors, for example environmental concerns, indicating the extent to which a person takes environmental issues seriously or worries about the current state of the issue. Finally, the framework of Massfeller et al., similarly to Huijts et al. (2012) also considers knowledge a crucial determinant – including, on the one hand, knowledge about the existence and details of the regulation itself, and on the other hand, content knowledge about sustainable farming practices.

Empirical research on agri-environmental schemes and farmers’ perception and behavior

The majority of existing empirical research on acceptance of and compliance with agri-environmental policies focuses on the analysis of voluntary agri-environmental schemes (AES). The concepts of acceptance, perception, and willingness to engage are often used interchangeably. In this regard, Schroeder et al. (2015) assessing farmers’ acceptance and perception of AES in Northern England through face-to-face interviews with farmers, find a very high (positive) impact of farmers’ families on the perception of AES. The analysis further finds that a high level of bureaucracy resulting in extensive paperwork reduces the acceptance of AES, as do concerns about production losses.

Mills et al. ((Mills, et al., 2017)) investigate both the farmers’ willingness and ability to engage with environmental

advice and to undertake environmental management more in general, through qualitative interviews with British farmers. They find that personal beliefs and moral norms are key explanatory factors, which can be influenced by the family members of farmers. The authors underline that pragmatic and economic factors drive farmers' engagement or non-engagement in environmental management practices, too. Moreover, they indicate the relevance of social influence of the farming community and the broader society on farmers' attitudes, while not considering the potential influence of institutional design of governance and management approaches on farmers' environmental (non-) engagement. Overall, they detect a high-level heterogeneity in the group of farmers regarding their level of environmental engagement.

Siebert et al. (2010) find that local advisors have an impact on the participation of farmers in environmental conservation measures, but state that financial gains, based on cost–benefit calculations, are the main reason for farmers to participate.

In a more recent study, Ptak et al. (2023) analyze the influence of agricultural advisors and the level of general political trust on farmers' compliance with regulations for reducing nitrate pollution from agricultural sources in Poland. Conducting a survey as well as interviews with farmers and authorities, the authors find that these two factors indeed play a significant role in explaining farmers' decision making. Ptak et al. (2023) do not consider the potential impact of further explanatory factors for (non-) compliance. In contrast to all studies presented above, Schroeder et al. (2015) find that the influence of advisors on farmers' acceptance of AES is rather low.

As one of fewer studies focusing on legally binding regulations against water pollution through agriculture, Winter and May (2001) analyze Danish farmers' compliance with different regulations. While putting a strong focus on the analysis of cost–benefit calculations, including the consideration of enforcement factors such as sanctions for explaining compliance, the authors find that social and individual normative factors such as a more general duty to comply are as relevant for compliance as cost–benefit calculations. The compliance rate with regulations as reported by the farmers is extremely high and might not be realistic, thus pointing to one of the major challenges of empirical compliance research, in particular in the context of legally-binding regulatory contexts, where reliably measuring actual compliance is often highly difficult (Carter and Siddiki, 202).

In a comprehensive empirical study conducting interviews with Chinese farmers and local experts, Yan et al. (2017) test the influence of eight potentially explanatory factors for compliance with legally binding pesticide regulations, as well as their interactions. These factors include

cost–benefit calculations, knowledge resources and further dimensions of individual capacity to comply, as well as social norms and morals. Results of their QCA-analysis indicate that the interaction of individual moral conceptions, social norms, and a positive cost–benefit analysis, in combination with either a high general duty to obey or a high perceived probability of being sanctioned are the most relevant explanations for compliance of farmers. In contrast to Winter and May, this study relies on the use of semi-structured interviews with farmers, arguing that this data source might be better suited for overcoming the complications of studying and measuring actual compliance.

While most of the existing empirical literature marginally considers policy design factors as determinants of acceptance of agri-environmental regulations, Brown et al. (2021) underline the high importance of different policy design factors for understanding farmers' decision making with reference to agri-environmental policies, in their critical analysis of the rationale of European agri-environmental subsidies. According to the authors, high policy complexity, inflexibility as well as administrative burdens negatively influence farmers' willingness and motivation to participate in AES.

Bringing together the conceptual-theoretical work on acceptance with empirical research insights from research on the determinants of acceptance of and compliance with AES by farmers, allows for developing a comprehensive conceptual framework of major determinants of policy acceptance. In the following, this framework is applied to systematically study farmers' acceptance of the fertilizer ordinance in Northern Germany by asking: What are central determinants of individual acceptance and acceptance deficits of the fertilizer ordinance by farmers? And: How do the different determinants interact with each other?

The framework assumes that individual acceptance of the ordinance depends on the interaction of five key explanatory determinants, which proved to be of high explanatory power according to existing research insights. These key explanatory determinants are economic, socio-political, and normative determinants, as well as policy design factors and knowledge. Each key explanatory determinant includes more specific explanatory factors, once again derived from existing evidence within previous conceptual and empirical research, so that the overall framework contains ten potential explanatory factors (see Fig. 1), which have been tested within the interviews with farmers, farmers' associations, and political authorities.¹

¹ As existing literature also indicates individual acceptance may furthermore depend on demographic factors as for example age and gender. Due to the small sample size is it not possible to study the influence of these factors, and they are therefore excluded from the conceptual framework for the empirical analysis for practical reasons, without contesting their relevance.

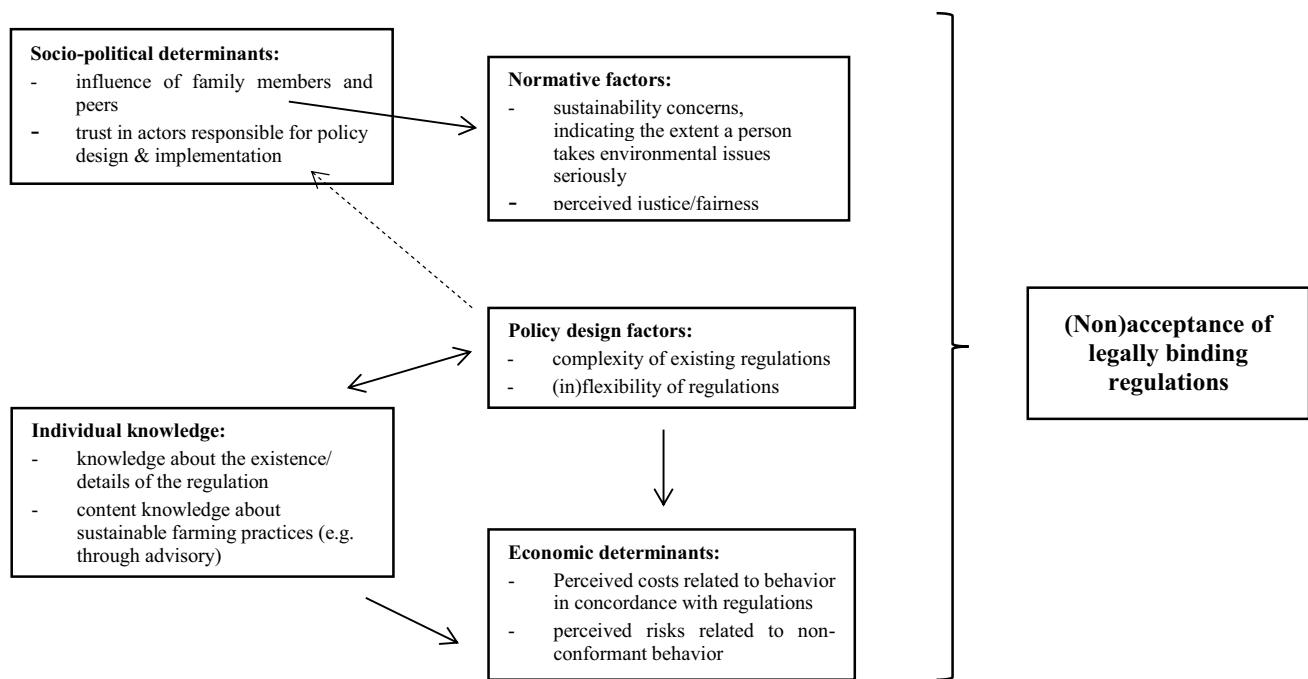


Fig. 1 Framework of interacting explanatory determinants of individual acceptance

The framework assumes that economic determinants influence individual acceptance on the one hand through perceived costs related to behavior in concordance with existing regulations, and on the other hand through perceived risks related to non-conformant behavior – in particular economic or other penalties.

Normative determinants refer in particular to sustainability concerns, indicating the extent to which a person takes environmental issues seriously in the context of his or her position on the value of good farming. These normative determinants can at least partly be affected by the social influence of family members and peers. Norms of justice may also be of explanatory relevance, either in the form of perceived fairness of the decision-making process of the regulations, or perceived fairness of the distribution of costs and benefits resulting from the implementation of regulations.

Knowledge on the other hand plays a role for individual acceptance in the sense of knowledge about the existence and details of the regulations themselves, and on the other hand content knowledge about sustainable farming practices. As such, knowledge is also interrelated with economic determinants, as it may influence the perceived costs of behavior in concordance with specific regulations. Relevant economic determinants also include perceived risks related to non-conformant behavior.

The framework also assumes that policy design factors matter, in particular as a high complexity of regulations and their inflexibility may reduce acceptance of regulations. In

this regard the existing governance structures, in particular different involved governance levels, in case of eutrophication governance from the local to the European level, as well as their interactions also have to be considered. Moreover, interaction effects with economic cost are likely to occur here as well. Finally, trust in actors responsible for policy design and implementation, a further socio-political determinant, also influences individual acceptance.

The case of the German fertilizer ordinance

Due to the significant impact of agricultural activities, especially fertilizing practices, on eutrophication, increasing regulative measures have been implemented to reduce nutrient input into marine waters. In the European context, a crucial regulation is the EU-Nitrates Directive adopted in December 1991. In Germany, the EU-Nitrates Directive has been implemented primarily in national regulations through the Fertilizer Ordinances. While the first German fertilizer ordinance was adopted in January 1996, there have been revisions in 2006, 2017, and 2021 – mainly due to failures to fulfill the obligations under the EU-Nitrate Directive. The national fertilizer ordinance sets basic legally-binding national standards for the use of nitrogen and phosphorus in agriculture with reference to absolute application limits, as well as periods and context conditions, under which fertilizer use is not permitted. Moreover, it regulates storage requirements for manure and digestates, and stipulates

Table 1 List of interviews and farm types

ID	Interview Date	Interviewee	Farm Type	Size of Farmed Land
IF1	17.01.2023	farm administrator	seed breeding farm	790 ha
IF2	17.01.2023	farmer	agricultural farming + some forestry	ns
IF3	06.02.2023	farm administrator	diary farming + agricultural farming	165 ha
IF4	08.02.2023	farm administrator	agricultural farming	1600 ha
IF5	21.02.2023	farm administrator	agricultural farming + biogas plant	1200 ha
IF6	27.02.2023	farmer	diary farm + biogas plant + some agricultural farming	650 ha
IF7	08.03.2023	farmer	primarily dairy farm	95 ha
IF8	14.03.2023	farmer	primarily dairy farm	275 ha
IF9	09.05.2023	farm administrator	agricultural farming	260 ha
IF10	22.05.2023	farmer	agricultural farming + biogas plant	320 ha
IF11	30.05.2023	farmer	agricultural farming + biogas plant	210 ha
IF12	31.05.2023	farmer	diary farming + pig fattening	210 ha
IF13	31.05.2023	farmer	agricultural farming + biogas plant	500 ha
IF14	12.07.2023	farmer	diary farming + some agricultural farming	145 ha
Interviews with political and administrative authorities and associations				
IA1	27.2.2023	representative of farmers' association		
IA2	28.02.2023	representative of chamber of agriculture of Schleswig-Holstein (SH)		
IA3	20.04.2023	3 representatives of German Farmers' Association in SH		
IA4	03.05.2023	representative of agricultural ministry in SH		
IA5	24.05.2023	representative of environmental ministry in SH		
IA6	02.06.2023	4 employees of farming advisory		

specific soil analysis. It thus provides the legal framework within which the federal states of Germany must operate. The federal state can, however, issue additional regulations to reflect regional conditions and challenges. They are moreover responsible for the implementation and enforcement of the specific regulations. The federal state of Schleswig-Holstein, relevant in the present case study, enacted its first fertilizer ordinance in 2018. The ordinance has been revised in 2020 and 2022. The latest version of the fertilizer ordinance comprises a baseline for the absolute amount of nitrate application (170 kg N/per hectare per year for organic nitrogen application), a mandatory nutrient advisory consultation, a mandatory fertilizer analysis before application, the stipulation of an immediate incorporation of fertilizer (within one hour), as well as several defaults for seasonal and soil-specific conditions of fertilizer uses. It also includes provisions for inspection and penalties for non-compliant behavior. In theory, penalties can be quite substantial, particularly in cases of repeated violations (up to 50 000 Euros per violation).

Research design and methods of the empirical analysis

Based on the ten interacting potential explanatory factors of the conceptual framework presented above, a semi-structured interview guideline was created, including 13 basic questions to gather detailed information on the explanatory factors, as well as their interaction dynamics. After several test-interviews with farmers and some minor revision of the interview guideline in December 2023, twenty semi-structured qualitative interviews were conducted between January and July 2023, fourteen with farmers in the region around the Eckernförde Bay, and six interviews with farmer's associations and political authorities all situated in the coastal federal state of Schleswig-Holstein (see Table 1).

Additionally, data material such as policy documents, and information provided by political authorities and ministries was analyzed. The purpose of this is twofold: to cross-check the interview findings and to complement incomplete information on specific issues discussed during the interviews.

The selection of interview partners was mainly based on a snowball system, meaning that each interviewee was asked for recommendations for further interview partners at the end of their interview (Merkens 2017). This of course comprises the risk of selection biases. For this reason, where feasible, an additional purposive sampling strategy (Flick 2007) was deployed, based on certain criteria of differences with regard to the characteristics of the farms. These criteria are: smaller and larger farms, agricultural farms and livestock farms, and family farms and farms managed by

administrators. Owing to the limited sample size, the results should not be interpreted as representative of Schleswig–Holstein or of regions beyond. They do primarily serve as a further qualitative differentiation of existing explanatory patterns within research on individual acceptance, both regarding the specification of the explanatory determinants, and a more comprehensive consideration of the interaction of different explanatory factors.

The interviews lasted for about an hour on average and were conducted within the farms of the interview partners. They were recorded and transcribed. Subsequent data evaluation was performed based on the qualitative content analysis according to Philipp Mayring (2011) and through the qualitative data analysis software MAXQDA.

The interviews were coded using a mixture of deductive and inductive approaches. Thus, codes basically follow the deductively derived ten explanatory factors (see Fig. 1). These codes were complemented with additional codes derived inductively from the data material (Mayring 2007), whenever a novel explanatory factor was mentioned in an interview. The interview material was revised several times to iteratively refine the coding (Ramsenthaler 2013). The coding unit used was one response by the interviewee (until interrupted by the interviewer for asking a different question or for commenting the response),² which seems most adequate to conceive one thematic unit of meaning (Hildebrandt et al. 2015).

Empirical findings

Most interviewees generally agreed with the necessity of regulating fertilizer uses in the agricultural sector. Nonetheless, they objected to certain provisions of the current fertilizer ordinance. At least some kind of critique was voiced in all 14 interviews with farmers. Some farmers principally endorsed the fertilizer ordinance by underlining for example that “on the whole it makes a lot of sense” (IF7) and only criticized specific provisions. This included, e.g., the *per se* prohibition of fertilization on frozen ground, as it offers high economic functionality. Some interviewees also voiced skepticism about the harmful effect of fertilization on frozen ground if the method is deployed in a skilled manner. One of the interviewees argued:

“But let’s say like ten days ago, two weeks ago, when we had light night frosts, where you could simply drive at night without it smearing, without damaging the surfaces, where they thaw again during the day.

That would be a classic phase where I would say that would actually be very justifiable from a functional and environmental perspective.”

The majority of interviewees, however, voiced more fundamental critique and highlighted a number of different objections to the fertilizer ordinance.

High relevance of economic determinants

Results of the qualitative content analysis of the interviews indicate that economic factors play an important role in explaining existing individual acceptance deficits. They were mentioned by all farmers to various extents, and confirmed in the conversations with political authorities. More specifically, interviewees criticized the fertilizer ordinance for reducing agricultural productivity due to harvest losses caused by changes in fertilizing practices. For example, one farmer noted: „But if we are not allowed to fertilize the wheat in the end, then we are missing this protein content and we only sell feed grain” (IF4), and another one underlined that due to the fertilizer ordinance: “of course, there is a great fear that you won’t be able to generate the profits anymore” (IF8). In two interviews, references to experiences in Denmark were made, where, in the past, rigid restrictions of fertilizer regulations allegedly contributed to a decrease in the protein content of wheat and thus to income losses of Danish farmers. Farmers’ association representatives similarly criticized potential productivity losses, opposing regulations on the overall quantity of fertilizer applied as well as specific rules governing fertilization practices, such as the ban on fertilizing frozen ground.

Moreover, respondents repeatedly highlighted the economic challenges faced by livestock farms under the fertilizer ordinance. The manure produced by livestock farms is typically applied as organic fertilizer, however, the ordinance restricts the amount of fertilizer that may be used. If livestock farmers do not operate biogas plants, they might have to transport the manure to other farms or regions with less life-stock farming, something which creates additional costs. To provide an example, one of the farmers said: “it must also be said quite clearly that the manure factor is really a pressing issue for some farms” (IF11). And another farmer even more clearly underlined: „On the one hand, they have their cattle in number, and then they have to be able to somehow spread their manure.” (IF14). Some interview partners also stressed that the problem of manure surplus in livestock farms is aggravated by an increasing regional concentration of livestock farms. Conversations with the political and administrative authorities confirmed that surplus manure in specific farms or areas is an important factor influencing attitudinal and behavioral dimensions

² The argument was coded when it occurred for the first time in one segment, following references were not coded separately but counted as accomplishment of the same argument.

of acceptance of the fertilizer ordinance. However, three of the interviewed livestock farmers had established private manure interchange formats with neighboring arable farms (where the receiving farmer bears the transportation costs but does not pay for the manure).

A further issue, the recent steep rise in mineral fertilizer prices, was not raised directly by interview guideline but emerged inductively. According to several interviewees, the increasing prices induce a more parsimonious use of mineral fertilizer. While the exact effect on behavior in concordance with the fertilizer ordinance is unclear (as reduced amounts of mineral fertilizer might go in parallel with increases in the use of organic fertilizer (manure)), several responses of farmers suggest a greater acceptance of this market-based mechanism compared with political interventions, such as regulations, which are less widely accepted. For instance, one of the interviewees stated: "I would say that the high prices of fertilizer in recent years – and today – are actually the best fertilizer regulation". Similarly, another farmer argued more generally that "any political intervention usually backfires. It is also past the need." (IF5).

Apart from perceived costs resulting from a behavior in concordance with the provisions of the fertilizer ordinance, farmers' behavioral (non-)acceptance may also be driven by the perceived risks related to non-conformant behavior. As outlined above, theoretically penalties for non-conformant behavior with the fertilizer ordinance can be significantly high, though actual controls are seemingly rather infrequent.

In the interviews, farmers were also asked about the frequency and overall perception of these controls. Most farmers state that they do not remember on-site controls for the fertilizer ordinance. Some farmers were also rather reluctant to respond to this question. Only one farmer stated that years ago, he had been controlled and had been moderately fined for non-compliance due to late fertilizer reporting. The interviews with administrative authorities tended to support the impression of infrequent (on-site) controls, it was however not possible to get access to consistent data on controls. So, while in summary it seems that the perceived (and actual) economic risk for non-conformant behavior with the fertilizer ordinance is rather low on side of the farmers, the reactions of several interviewees also underline the challenges of gathering data on non-conformant behavior in the context of legally binding regulations.

The influence of policy design factors

Apart from economic factors, findings indicate that the policy design of the fertilizer ordinance also has a strong explanatory power regarding its acceptance or acceptance deficits. Three major points of critique emerged in the

interviews: over-complexity, lack of practicability, and planing insecurity.

Regarding over-complexity, one farmer for example argued:

„a problem with this system is that it is so complicated [...] But if anyone could tell me all the parameters of the current fertilizer ordinance and what distances³ and where and how ... No-one can remember that anymore" (IF6).

Secondly, a recurring and frequently voiced concern across all interviews with farmers also was the inflexibility of the regulations and their resulting lack of practicability. In particular, it was criticized that the heterogeneity of farm types and further context conditions had not sufficiently been taken into consideration within the regulations. Some statements explicitly referred to differing conditions between geographical regions, others were more general, such as: „such rules are made with a watering can and are not properly differentiated “ (IF7). Another farmer criticized that acting in concordance with the provisions of the fertilizer ordinance would not leave him any room for maneuver, albeit such flexibility is highly important in farming due to unpredictable and changing weather conditions (IF4).

A related issue, which was strongly criticized by numerous farmers, has been the rigidity of documentation duties of fertilizer use. For instance, one farmer stated: „That's the most annoying thing. I can behave in an exemplary manner [but] if I did not submit the report in time, I did it wrong" (IF11).

Interestingly, the criticisms of policy design were often closely connected to another issue, that of farmers perceiving regulating authorities and/or politicians as questioning the farmers' competence. As an example, one farmer stated: “And these things, I know that, are designed to tell me, dear [name of farmer], you don't know how to do it right anyway.” (IF11). It can thus be argued that the non-acceptance of the policy design of the regulations interacts with issues of political (mis-)trust.

A third group of arguments is centered around the lack of planning security due to frequent regulation changes, which was articulated numerous times in the interviews with farmers. In this regard, one interviewee argued: „It bothers me massively that we can't even have something that lasts three or five years" (IF6), and another one said: “planning security has deteriorated in the sense you don't know how long political decisions will last” (IF5). According to the farmers'

³ The term of „distances “ is referring to the provisions of the fertilizer ordinance with regard to the extent of so-called buffer zones, where, depending on the degree of slope inclination, the use of fertilizer is completely forbidden.

statements, this lacking planning security leads to new economic challenges by increasing uncertainty about harvest income.

The impact of knowledge factors

Regarding the determinant of knowledge about the specific provisions of the fertilizer ordinance, most of the interviewed farmers knew the details of these provisions very well. Only a minority of farmers voiced problems of understanding the ordinance due to its (perceived) over-complexity (see also findings on the influence of policy design). Thus, lack of knowledge regarding the fertilizer ordinance is unlikely to be a main cause of acceptance deficits.

As empirical research has stressed the importance of advisors in increasing acceptance of regulations by diffusing knowledge on sustainable farming practices (Siebert et al. 2006; Siebert et al. 2010; Ptak et al. 2023), the interview guideline also asked for the existence and subject area of external advisory in the context of the implementation of fertilizer ordinance. In general, external advisory was considered highly important by most of the interviewed farmers. Most farmers make use of different, partly private advisory agencies, except from administered farms. The advisory system is, however, highly diverse, ranging from cattle specialist advisory over business consultancy to specialized agrarian advisory services.⁴

An increasing leverage for advisors to influence the farmers' knowledge about and perspective on the fertilizer ordinance can be assumed, but the prioritization of sustainable farming practices may very much depend on the specific type of advisory.

The influence of socio-political determinants

As far as the socio-political factors are concerned, the interviews revealed that it is political trust⁵ in particular, which influences the acceptance of the fertilizer ordinance. In all interviews with farmers, mistrust towards individual political actors or political parties, considered to be responsible for the policy design and/or implementation of the fertilizer ordinance, was propounded. A statement exemplifying political mistrust quite well, is the following:

"In these parliaments there are a lot of people who are not from agriculture and who are somehow civil servants, an incredible number of lawyers and then also teachers, who of course have the opportunity to say with relative certainty, well, then let's go into politics, because I am a civil servant and I can always return to this profession after ten years." (IF2).

Other statements by farmers underlined a deep mistrust in politicians, based on a perceived distance and missing political interest in agricultural issues. Expressed mistrust was especially high towards the Green Party and its representatives, who are currently part of a coalition government in Schleswig-Holstein and to whom the perceived deficits of the fertilizer ordinance were specifically attributed.

Some farmers indicated that this casts the other coalition partner, the Conservative Party, in a negative light. For instance, one respondent stated that he cannot forgive the current governor in Schleswig-Holstein from the Conservative Party, for "betraying the farmers" by entering into a political coalition with the Green Party (IF 10).

While political representatives of the Green Party were predominantly mistrusted and perceived as detached from agricultural practices, interestingly one political actor of the same party, Robert Habeck, was mentioned positively several times despite his party affiliation. One of the interviewees for example stressed: "a big, big advantage of Mr. Habeck, is that he is a person, who really looked at things and implemented them relatively quickly and said, that's one thing, we all have to work on" (IF11). Similarly, another politician (from the Free Democratic Party) was also regarded as more trustworthy than other political actors, being perceived as attentive to and communicative with the farmers.

Elevated mistrust towards political actors also seems to contribute to the farmers' perception that politicians have little trust in farmers' commitment to environmental issues and in their professional expertise in general, as articulated in several interviews (see also findings on the determinant of policy design). Farmers felt mistrusted not only by politicians, but also by specific groups of society, first and foremost environmental organizations and the media. In turn, the interviewees reproached the media for disinformation and negative tabloid journalism.

While several interview partners underlined that this perceived societal mistrust in farmers affects them negatively,⁶ it was hardly traceable in how far this mistrust affects the acceptance of the fertilizer ordinance concretely. Only in

⁴ During the conversations with political authorities, the increasing demand for advisory formats was confirmed as well.

⁵ Following Huijts et al. (2012), trust is considered to be an intuitive affective notion towards an entity. Political trust consequently refers to affective notions towards regulatory authority. While operationalizations of regulatory authority differ within existing research, standard measures in comparative political research usually focus on institutions such as political parties, politicians, governments and parliaments, and the police (see Marien et al., 2019).

⁶ One farmer revealed: „That we are always pilloried. We are always to blame for everything. At first, I thought this isn't so hard. And if you want to become a farmer, you just do it and then you will manage it somehow. But I've noticed lately that this is a central issue." (IF14).

one interview with a representative of the German Farmers' Association, a potential link towards increased behavioral acceptance can be derived through the following argument referring to negative media presentation of farmers due to irregular fertilization practices: "We have to get out of the headlines and get out of the bad ones that exist at the moment" (IF14). In theory, the effect of the perceived mistrust may be twofold: farmers might either feel increased psychological pressure to act in concordance with regulations, or might feel like "scapegoats" for everything, regardless of how they actually behave, and tend to affectively react with reduced acceptance of regulations.

Normative determinants of (non-)acceptance

Finally, beliefs about general normative guidelines for agriculture also seem to influence the acceptance of the fertilizer ordinance. Most interviewed farmers saw themselves first and foremost as economic entrepreneurs, whose priority is economic efficiency. Expressed in the words of one farmer: „first of all, in the end it has to be said, we simply put profitability first, that simply has to be given" (IF6), or of another: "Output, output in the form of grain or pork or beef or milk or per input perhaps? How efficiently do I use the resources" (IF1). Beyond that, the normative importance of sustainable agricultural production was stressed repeatedly by the majority of farmers. Several interviewees also linked the concept of sustainability specifically to environmental and nature protection.⁷ Concerns about the protection of the marine ecosystem was, however, not explicitly mentioned by any farmer.⁸ Moreover, in most interviews, the meaning of sustainability generally remained rather vague. Still, what can be derived from the interviews, is that the concept of sustainability of most farmers implies concerns about long-term future agricultural productivity, and that this is related to the importance dedicated to the family and the principle of family business. One farmer explained: "So, the main guideline, as stupid as it sounds, is sustainability, because I want to hand over my business to the next generation if possible" (IF13). Another farmer explicitly stated: „compared to many other industries, [...] we have focused on sustainability and also on long-term effects, because most farms are still family businesses" (IF9). In this sense, most interview partners expressed a responsibility towards the next generation of making efforts towards fostering sustainability in order to leave them with sound options for

the future and of refusing economic strategies focusing on maximizing short-term profits at the expense of ecosystem protection.

A further normative issue, potentially relevant for acceptance or deficits of acceptance of fertilizer regulations are concerns about justice and fair treatment. The interviews contain numerous references to a perceived "unfair treatment" either by comparison with other European countries or by comparing different groups of polluters within Germany. So, on the one hand, it was repeatedly put forward that farmers in Germany were discriminated against, compared to other EU countries, where fertilizer regulations and their control was said to be handled less strictly. On the other hand, various interview partners voiced that they felt treated unfairly because of increasingly severe regulations, which are restricting agricultural productions, while for example sewage plants continue to foster eutrophication. One farmer stated:

„But from our point of view, the topic is actually not treated fairly, so that you can really say, where do things [nitrogen] come from? And that you can really say, for example, 20% agriculture, 40% sewage treatment plants" (IF6).

Sewage treatment plants were also mentioned by another farmer interviewee: „As far as sewage treatment plants are concerned, of course, I definitely think that there is a joint responsibility [for nutrient input in (marine) waters" (IF12). This perception of unfair treatment may be linked to the factor of lacking political trust in that the low level of political trust described above fosters the perception of unfair treatment by political authorities.

Discussion of results

Findings of the empirical analysis highlight the prominent role of economic factors for explaining acceptance and acceptance deficits of the fertilizer ordinance. In particular, perceived costs (either harvest losses or transportation/sales cost for manure) resulting from a behavior in concordance with the provisions of the fertilizer ordinance influence farmers' non-acceptance of parts of or the entire fertilizer ordinance. Moreover, farmers' behavioral (non-)acceptance may partly also be driven by relatively low perceived risks related to non-compliant behavior. Gathered empirical insights on non-conformance are, however, rather incomplete so that more reliable assertions are not possible.

Findings on the high relevance of economic factors are mostly in line with previous research, underlining differing cost-benefit calculations in livestock farms as compared to

⁷ For example, one farmer said: "[To be economically well positioned] and also to perform well in terms of the environment and nature conservation." (IF3).

⁸ However, one farmer mentioned the importance of water protection, referring to drinking water by arguing: „We all want to preserve nature and ensure that drinking water continues to be drinkable " (IF5).

arable farming (Winter and May, 2001) as well as calculations of financial gains or losses in general (Siebert et al. 2010; Schroeder et al. 2015). An interesting finding regarding the policy relevance of the empirical insights is that costs of behavior in concordance with the provisions of the fertilizer ordinance in live-stock farms can be minimized for example through the establishment of private manure interchange formats with neighboring arable farms. Such manure interchange formats are, however, not feasible everywhere and they are further challenged by the increasing industrialization of agriculture in the sense of an agglomeration of livestock farms in specific regions as well as the current trend towards increasing sizes of farms.

Not necessarily related to behavior in concordance with the fertilizer ordinance, but of general political relevance, is the factor of sharply increased mineral fertilizer prices and reductions in (mineral) fertilizing practices. Notably, several farmers' statements suggest that attitudinal acceptance of market-based mechanisms is higher than that of political interventions. This phenomenon might in turn be influenced by the ascertained low level of political trust but also by normative self-concepts. While these insights are preliminary assumptions only, it might be highly interesting to further explore this issue in future research.

Another crucial insight of this analysis is the high relevance of individual political trust/mistrust for explaining acceptance of fertilizer ordinance and acceptance deficits. The impact of different dimensions of trust on acceptance has been confirmed by previous studies. Ptak et al. (2023) identify a low level of trust of Polish farmers in political institutions, which according to the authors may ultimately hamper implementation efforts of nitrate pollution governance (Ptak et al. 2023). Blackstock et al. (2010) more specifically find that trust in consultants matters for the implementation of nutrient management regulations by farmers (Blackstock, et al., 2010). While the level of trust in particular consultants has not been tested in this paper, findings indicate that first of all there actually is a lot of mistrust in political actors and political parties responsible for the policy design of the fertilizer ordinance from part of the interviewed farmers. This mistrust is mostly based on a perceived "distance" and missing political interest in agricultural issues of the politicians. Moreover, findings seem to indicate that this mistrust contributes to acceptance deficits of the fertilizer's ordinance, and third, that low political trust of farmers indirectly also seems to foster the farmers' sensation of a low trust of politicians towards the professional expertise of farmers. In this sense several farmers for example ascribed perceived deficits of the policy design of the fertilizer ordinance (in particular the perceived high rigidity) to lacking trust in farmers' competence.

This is of particular importance as by this ascription farmers create a complex and interconnected "wicked problem" (Jentoft and Chuenpagdee, 2009; Lawrence et al., 2022). Wicked problems are difficult to capture and tackle, in particular because there are varying sensations, knowledge, and interests involved, and thus for example no straight forward true or false solutions or answers to these kinds of problems are possible (Lawrence et al., 2022; Grünhagen et al., 2022). The specific challenge of such wicked problems in the given context has also been confirmed in interviews with advising agricultural authorities commenting on the problem of political mistrust by saying: "Then it is sometimes difficult to reach such people again through advice" (IA6).

Low trust in politicians might also contribute to explain the general skepticism towards political intervention, as compared to market-based regulative approaches, which has been highlighted above in the context of increased prices for mineral fertilizer. On the other hand, there have also been a few statements by farmers hinting towards the ability of certain political actors to breach political mistrust when being perceived as a practitioner-oriented individual attentive to the farmers. Here again it would be interesting to analyze this issue more comprehensively in future research.

Apart from economic factors and political (mis)trust, this analysis also underlines the high relevance of factors of policy design for individual acceptance of regulations. This finding confirms Schroeder et al. (2015), stressing high load of paperwork as an important factor for farmers' non-acceptance of environmental conservation measures. Due to the qualitative inductive coding of the interview data, a further differentiation of the specific issues causing deficits of acceptance was possible. This study found that, firstly, farmers most frequently criticized the lack of flexibility and practicability of the regulations. A second major point of criticism is the (perceived) over-complexity of the fertilizer ordinance, causing actual problems of understanding. A third critique relates to lacking planning security due to frequent changes in the institutional design of regulations. These frequent changes were specific in the country case of Germany, where there have been disproportionally many reforms of the federal and state fertilizer ordinances due to repetitive criticism and legal action from the European Commission and the Court of Justice, respectively, regarding inadequate implementation of the EU Nitrate Directive. After the cessation of the EU infringement proceedings against Germany in 2023, the frequency of reforms of the fertilizer ordinances are expected to decline.

The analysis also further disentangled the influence of the factor of knowledge provision of advisory on acceptance of the fertilizer ordinance, which has been repeatedly considered in previous empirical research on AES (Blackstock et al., 2010; Siebert et al. 2010; Ptak et al. 2023). Regarding

the influence of advisory services, it can be highlighted that there is a growing importance of external advisory. In general, this opens more leverage for influencing farmers' knowledge about and potentially acceptance of regulations. Whether current advisory services have put a strong emphasis on sustainable farming practices could not be detected in detail and would require additional research in the future.

Results of the paper, moreover, clearly show that the farmers' families are an important influence regarding normative motivations of sustainable farming, which might indirectly also have consequences for the acceptance of the fertilizer ordinance. The finding that family exerts an important influence on farmers' perceptions and behavior has also been highlighted by other studies (Schroeder et al. 2015; Zebrowski et al., 2023). In contrast to the results of Schroeder et al. (2015), in the present paper family influence is mainly discernible indirectly through concerns of farmers about the wellbeing of children and/or future generations, in the context of beliefs about normative guidelines for farming. Most farmers appear to show sensitivity towards the acceptance of environmental protection especially based on the value of personal family legacies – albeit predominantly only if not at the expense of agricultural effectiveness.

Despite repeated distinct references to the problem of eutrophication of the Baltic Sea within the interview guide, voiced concerns about environmental protection did not explicitly refer to the protection of the marine environment, but remained rather general, with occasional specific focus on terrestrial environmental protection. While this issue, including reasons behind it, has not been further pursued empirically in the interviews, this might be an interesting question for future research. In particular, because a similar kind of “sea-blindness” (Feldt 2013) or land-sea divide (see for example Kuhn et al. 2024) has been detected by other studies as well – not only with regard to farmers but as a general phenomenon in society, only slowly changing more recently due to new (European) governance approaches (Morf et al. 2022).

Finally, the relevance of concerns about justice and fair treatment for acceptance and deficits of acceptance of the fertilizer ordinance resonates within the above presented conceptual research on acceptance (Lind et al. 2016; Huijts et al. 2012; Schade and Schlag, 2003; Tyler 1990; Van den Bos et al. 2014). This research has demonstrated that actors who perceive the distribution of costs and benefits resulting from the implementation of regulations as unjust are less likely to accept relevant regulations. Insights from this conceptual research have, however, to the author's knowledge, hardly been considered in empirical research on the individual acceptance of agricultural environmental regulations so far. Findings of the present paper may thus constitute a valuable contribution to further research in this field.

Conclusions

Based on an interdisciplinary research review of conceptual-theoretical work on acceptance and empirical research insights from research on the determinants of acceptance of and compliance with agri-environmental schemes by farmers, this paper has innovatively elaborated a comprehensive conceptual framework for the systematic empirical analysis of central determinants of individual acceptance and acceptance deficits of the fertilizer ordinance in Germany—including a detailed study of interacting mechanisms between the different factors.

Findings underline that a complex interaction of economic factors, policy design, as well as socio-political and normative factors explain individual acceptance and acceptance deficits. This study offers new insights on how the interacting determinants of economic concerns about reduced agricultural productivity, individual political mistrust, and policy design drive substantial acceptance deficits of the fertilizer ordinance. Of particular relevance seems to be an interacting mechanism, which so far has received little attention in existing empirical research: the entanglement of issues of political mistrust and policy design resulting in wicked problems, which are generally considered to be highly difficult to tackle (Lawrence et al., 2022).

Regarding the policy relevance of the results, the importance of factors such as political (mis-)trust and multi-level policy design of regulations, perceived as over-complex, impractical, and inflexible, may speak in favor of an expansion of (voluntary) governance approaches focusing on dialogue and exchange formats between farmers and political authorities, as well as an increase of state-run long-term advisory formats.

To ensure personalized trust-building, such approaches should be of local or regional character. In particular, as national (and all the more EU-wide) regulations are only to a limited degree able to factor in farm heterogeneity, resulting in regulations with higher inflexibility.

Considering that insights also indicate a high relevance of economic determinants, these approaches should not replace regulatory binding requirements, but solely complement them. The demonstrated wicked problem complexes can be addressed comprehensively only through such additional governance approaches. In doing so, and as already indicated by others (see for example Mills et al. 2017), differing governance and advisory approaches are needed for increasing acceptance of regulations and voluntary initiatives, depending on the different context conditions for example in life stock farms on the one hand and arable farms on the other hand or in family farms versus farms managed by external administrators.

Due to the small sample size, qualitative interrogation approaches, as well as potential selection biases regarding the interview partners, results of this empirical analysis of this paper can be generalized only to a very limited extent. However, this analysis has generated valuable new insights, whose more systematic relevance for explaining individual acceptance as well as interacting mechanisms can be tested and consolidated in further comparative case studies with large sample sizes and by drawing on more standardized quantitative survey data. Insights may also be relevant beyond the specific case of eutrophication governance and its acceptance, regarding the general role of agriculture for marine environmental protection and restoration, for example in issue areas of land-sea interaction such as the governance of pesticides, heavy metals, or the restoration of salt marches. Moreover, the relevance of explanatory factors, which proved to be rather independent of the specific context of agricultural production, such as political trust, perceptions of fairness, and policy design may also provide new insights for future research on marine environmental governance and its societal acceptance beyond research on agriculture and (marine) environmental protection.

Finally, an issue, which has been discernible on the brink of some of the interviews, are potential contradictions between environmentally sustainable agricultural practices at the farm level and global sustainability, for example by the externalization of ecological burdens to other regions. While a more detailed analysis of these contradictions has been beyond the scope of this paper, it merits more in-depth attention in future research.

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Declarations

Conflict of interest The author states that there is no conflict of interest.

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